RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME					
CENTRE NUMBER		INDEX NUMBER		CLASS	

H1 BIOLOGY **8875/02**

Paper 2 **14 September 2011**

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.
Circle the question number in the table on the right.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 10
2	/ 12
3	/ 7
4	/ 11
5 or 6*	/ 20
Total	**/ 60**

This Question Paper consists of **13** printed pages.

Section A

Answer **all** the questions in this section.

1. **PART I**

Fig. 1.1 is a high power electronmicrograph of a cell.

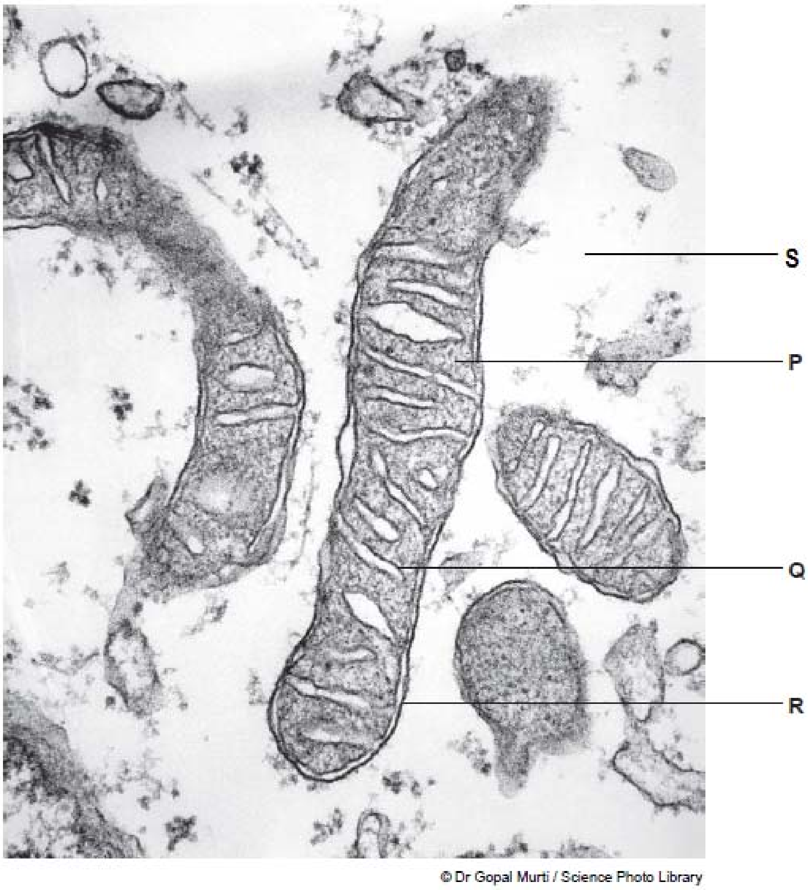

Fig. 1.1

(a) With reference to **Fig. 1.1**, where in the cell does each of the following occur? Indicate your answer with the appropriate letter. [2]

(i) *Krebs cycle* **P ;;**

Oxidative phosphorylation **Q ;;**

(ii) There are differences in the ways in which ATP is made in glycolysis and in the electron transport chain.

Give **one** of these differences. [2]

1. **In glycolysis no oxygen required/ direct link to reaction;**

   A: substrate level phosphorylation

2. **In electron transport oxygen required/ linked to electron carriers/ electrons from reduced coenzyme;**

   A: oxidative phosphorylation

1.

PART II

Lipid vesicles (i.e. spherical lipid bilayers) that contain Na^+-K^+ pumps as the sole membrane protein were prepared. For the sake of simplicity, assume that each pump transports one Na^+ one way and one K^+ the other way in each pumping cycle as shown in **Fig. 1.2**. Each Na^+-K^+ pump is oriented so that the portion of the molecule that normally faces the cytosol faces the outside of the vesicle.

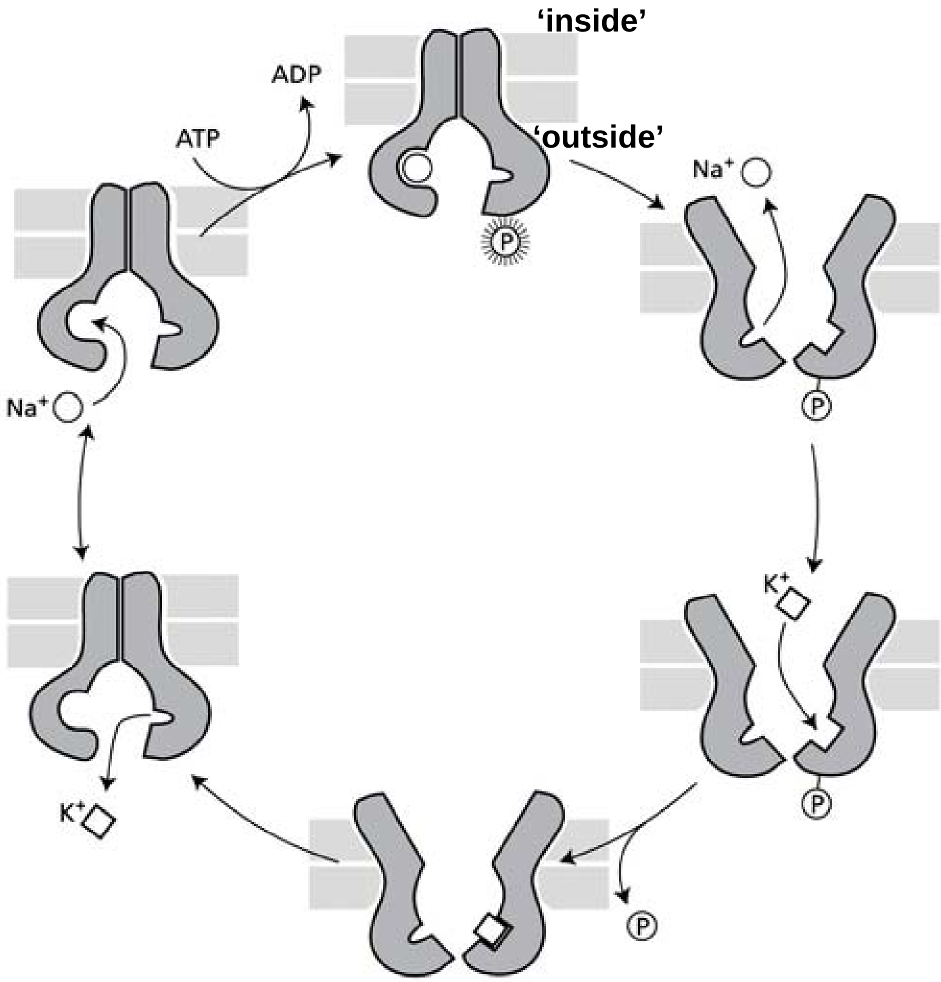

Fig. 1.2

With reference to **Fig. 1.2**, explain what would happen under each of the following conditions.

(b) The solution inside and outside the vesicles contains both Na^+ and K^+ ions but no ATP. [1]

1. **no ions will be pumped in the absence of ATP;;**

1. **(c)** The solution inside the vesicles contains both Na^+ and K^+ ions; the solution outside contains both ions, as well as ATP. [2]

1. **The pumps will hydrolyse ATP;**
2. **for the transport of Na^+ into the vesicles and K^+ out of the vesicles;**
3. **This generates oppositely oriented concentration gradients;**
4. **of Na^+ and K^+ across the membrane;**
5. **The pumps will stop; when the ATP runs out;**

(d) The solution inside contains Na^+; the solution outside contains Na^+ and ATP. [3]

1. **The pump will bind Na^+; hydrolyse ATP and transfer P_i to itself; and transport Na+ against concentration gradient;**
2. **The pump will then stop; because (i) K^+ is absent; and (ii) the conformational changes are strictly sequential;**

[Total: 10]

2. **PART I**

Fig. 2.1 is a diagrammatic representation of the process of transcription.

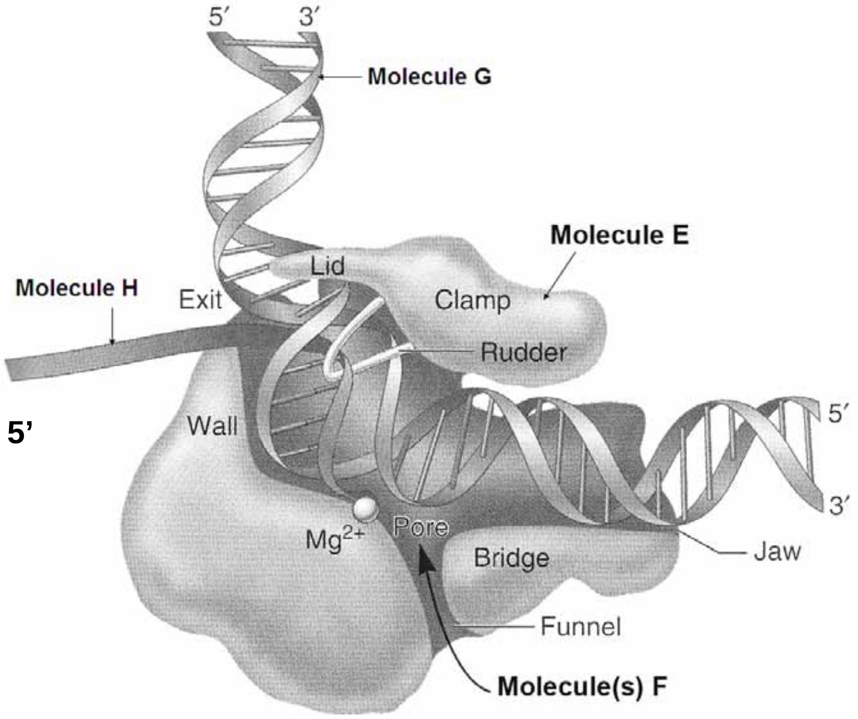

Fig. 2.1

(a) Describe how the structure of molecule **E** is adapted to its role in transcription. [2]

1. **molecule E has a <u>specific active site</u>;**

   Accept: active site being complementary to DNA

2. **<u>contact amino acid residues</u>; that <u>binds substrates</u> such as DNA template, ribonucleotides; and <u>nascent products</u> such as the growing RNA strand;**

3. **<u>catalyatic amino acid residues</u>; that <u>catalyses phosphoester bond formation</u> between growing RNA strand and incoming free nucleotide;**

4. **<u>funnel</u> allows <u>entry of nucleoside triphosphates</u>;**

5. **<u>rudder</u> <u>separates the RNA-DNA hybrid</u>;**

(b) Describe two ways by which the structure of molecule(s) **F**, which is / are adapted for transcription, differs from that of amino acids, which are adapted for translation. [2]

1. **a molecule of F (dNTP) is made of a central deoxyribose attached to <u>phosphate group</u> and <u>nitrogenous bases</u>; but an amino acid is made of a central α carbon attached to <u>amino group</u> and <u>carboxyl group</u>;**

2. **identity of molecule F is determined by <u>identity of nitrogenous base</u>; but identity of amino acid is determined by <u>identity of R-group</u>;**

2. (c) Using evidence from **Fig. 2.1**, explain why the process of transcription is conservative. [2]

1. **DNA double helix unwind and separate; to reveal the template strand;**
2. **after which both DNA strands rewind; to restore the double helix structure;**

(d) Explain why molecule **H** is shorter than molecule **G**. [2]

1. ***idea that* only copies one, gene / section / part / AW, (of DNA);;**
2. ***idea that* DNA comprises many, genes / alleles;;**

(e) Molecule **E** is inhibited by the toxin, α-amanitin. **Fig. 2.2** shows the effect when α-amanitin attaches to this molecule.

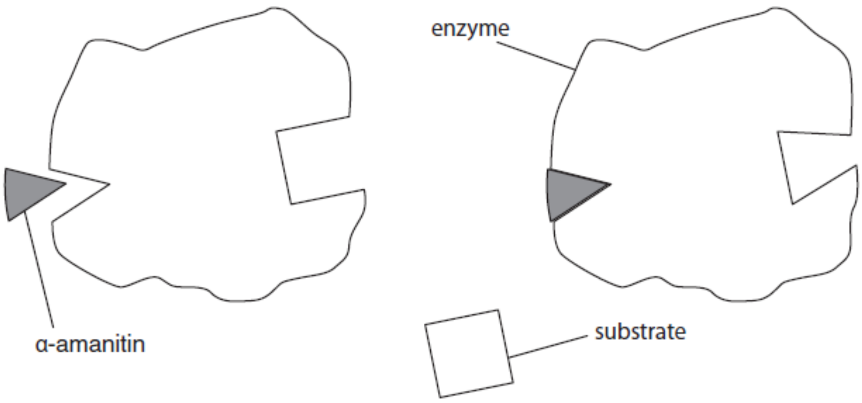

Fig. 2.2

(i) Explain how α-amanitin stops the formation of an enzyme-substrate complex during transcription. [2]

1. **non-competitive (inhibitor);**
2. **(α-amanitin / inhibitor / toxin) fits into, a place other than active site;**
3. **active site changes, shape / conformation / structure;**
4. **substrate no longer, fits / complementary to, active site;**

2. **(e)** **(ii)** The Roman Emperor Claudius was poisoned by his wife Agrippina when she gave him death cap fungus to eat. The death cap fungus contains α-amanitin.

Suggest how the toxin α-amanitin may lead to the death of an organism. [2]

1. **inhibits production of mRNA / mRNA not produced;**
2. **prevents protein synthesis / AW;**
3. **e.g. of, specific named protein / (vital) process, that may be affected;**

[Total: 12]

3. In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

Fig. 3.1 shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.

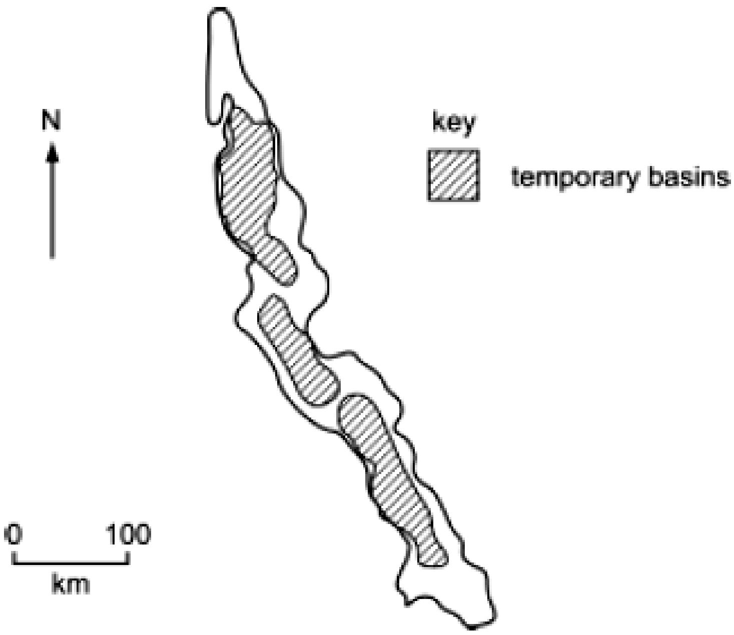

Fig. 3.1

(a) Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation. [3]

1. **variations in the population ; due to mutation/ different alleles;**
2. **different parts of the lake exerts different selection pressure;**
3. **favouring different phenotypes;**
4. **organisms with best-suited/favourable phenotype, survive and reproduce;**
5. **passing on their genes/alleles to the next generation;**
6. **sympatric/ allopatric/ geographical isolation/ accept hundreds of km apart/ no gene flow between different populations;**

Recent research has compared DNA sequence of cytochrome b gene of the six different species of *Tropheus* (**Fig. 3.2**).

```
T. brichardi     CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
T. kasabae       CTTTTAACTATGATAATAGCTTTTGTGGGCTACG
T. annectens     CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
T. moorii        CTTTTAACTATGATAACAGCTTTTGTGGGTTACG
T. polli         CTTTTAACTATAATAACAGCTTTTGTGGGCTACG
T. duboisi       CTTTTAACCATGATAACGGCTTTTGTAGGCTATG
                 ******** ** ****  ******** ** ** *
```

Fig. 3.2 shows part of the cytochrome b DNA sequence

(b) With reference to **Fig. 3.2**,

(i) explain why different species of *Tropheus* can have similar DNA sequence in their cytochrome b gene. [2]

1. **they might have <u>evolved from a common ancestor</u>;;**
2. **code for important <u>protein involved respiration</u>;;**
3. **any mutation will result in non-functional protein which will be fatal to the organism;;**

(ii) explain the advantage of the use of DNA sequence over amino acid sequence in determining the evolutionary relationship of *Tropheus* species. [2]

1. **change in DNA nucleotide sequence, does not change the type of amino acid;;**
2. **ref degeneracy of genetic code;;**

[Total: 7]

4.

PART I

Many bacteria make enzymes which protect the bacteria by breaking down the DNA molecules carried into the cell by viruses. The DNA target sites of four restriction enzymes are shown in **Table 4.1**. The points at which the strands of DNA are cut are shown by arrows and lines.

Table 4.1

restriction enzyme	target site
Sau3AI	↓G – A – T – C – – C – T – A – G↑
BamHI	– G↓G – A – T – C – C – – C – C – T – A – G↑G –
HinfI	– G↓A – N – T – C – – C – T – N – A↑G – 'N/N' may be any complementary base pair

(a) Suggest why these enzymes do not cut bacterial DNA. [1]

Selective addition of methyl ($-CH_3$) groups to adenines or cytosines within the sequence recognized by the restriction enzymes (restriction site);;

(b) With reference to the information in **Table 4.1**,

(i) explain whether or not a piece of DNA cut by ***Sau*3AI** could join with one cut by ***Bam*HI**; [2]

1. **yes;**
2. **same sticky ends/ GATC or CTAG;**
3. **complementary (bases);**
4. **hydrogen bond;**

(ii) show on **Fig. 4.1** the result of exposing this piece of DNA to ***Hin*fI**.

– G – A – T – T – C – A – G – A – A – T – T – T – C – G – A – A – T – C
– C – T – A – A – G – T – C – T – T – A – A – A – G – C – T – T – A – G

Fig. 4.1

two correct cuts;;

G| A T T C A G A A T T T C G| A A T C
C T A A |G T C T T A A A G C T T A |G

[1]

PART II

Fig. 4.2 shows a vector map of the plasmid vector pSING 003, which is used to clone the gene coding for human insulin.

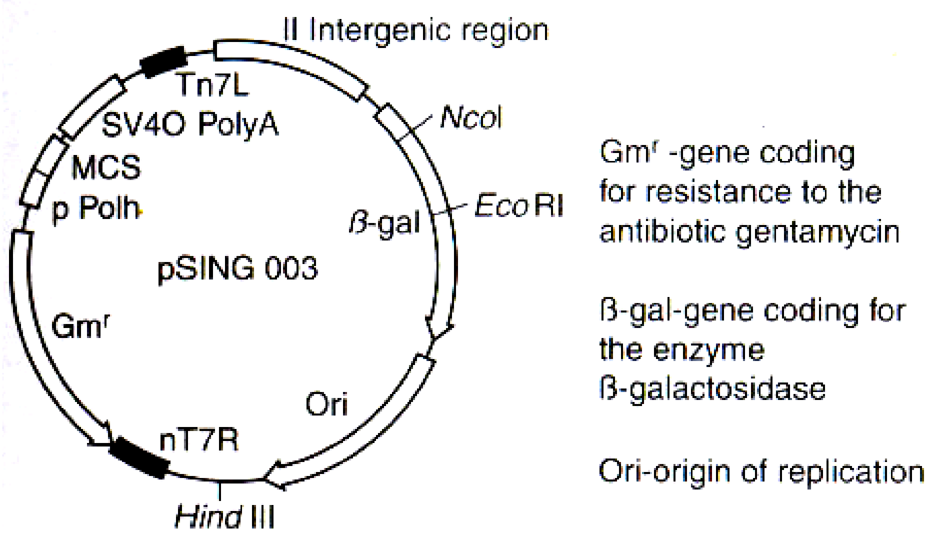

Fig. 4.2

(c) Describe, with reference to specific restriction enzymes, how the insulin gene can be inserted into the vector pSING 003. [3]

1. ***Eco*RI or *Nco*I used to cut; region flanking insulin gene;**
2. **same restriction enzyme (*Eco*RI or *Nco*I) used to cut; pSING 003 at β-galactosidase marker gene;**
3. **sticky ends; anneal by complementary base-pairing;**
4. **DNA ligase; added to covalently join the annealed DNA strands;**

(d) **Fig. 4.3** shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

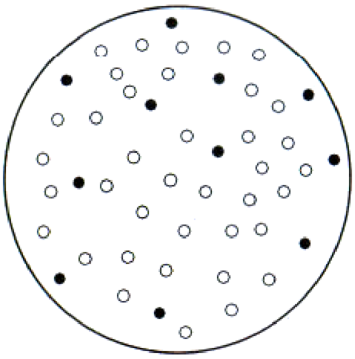

Fig. 4.3

(i) Explain why some colonies appeared white while others appeared blue. [2]

Colonies appear blue because

1. **these are the cells that contain the reannealed plasmid /without the insulin gene inserted;**
2. **can thus produce the functional β-galactosidase protein;**
3. **to catalyse conversion of the X-gal into a blue compound;**

Colonies appear white because

4. **insertion of insulin gene be inserted into the vector disrupts the nucleotide sequence of the β-gal gene;**
5. **cannot produce the functional β-galactosidase protein;**

(ii) Suggest why replica plating was not necessary in this experiment. [2]

1. **selection of transformed bacterial cells with recombinant plasmid;**
2. **can be done on <u>one</u> agar plate supplied with gentamycin and X-gal;**
3. **positive colonies (i.e. transformed with recombinant plasmid) would be gentamycin resistant; and appear white;**

OR

1. **replica plating is done only when selection cannot be done on one plate;**
2. **in cases where cells with the recombinant plasmid can <u>survive on one medium but do not survive on the other</u>;**
3. **the selective agents (e.g. two antibiotics) cannot be added to the same plate;**

[Total: 11]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

5. **(a)** Explain how the structure and properties of triglycerides are related to their functions in living organisms. [6]

Function		Structure-property
Storage of energy	**A1**	• large and uncharged; • insoluble in water; • no effect on the water potential of the cells; • prevented from leaving the cells;
	A2	• contain a larger number of hydrogen atoms per gram; • yields a greater amount of energy; • useful for animals in which locomotion requires the mass to be kept to a minimum;
	A3	• oxidised only after carbohydrates are depleted; • long-term energy store → important to hibernating animals;
Source of metabolic water	**A4**	• 2-fold more hydrogen atoms ; • yield water on oxidation during respiration; • a good source of metabolic water → important to desert animals;
Insulator	**A5**	• good thermal insulator/poor conductor of heat; • preventing excessive heat loss;
	A6	• electrical insulators; • components of the myelin sheath of nerve cells, allows for rapid transmission of impulse along myelinated neurone;
Buoyancy in animals	**A7**	• less dense than water; • providing aquatic mammals buoyancy;
	A8	• AVP ;;

5. **(b)** Explain the role of membranes in photosynthesis and respiration. [8]

Photosynthesis

B1 Structure: thylakoid membrane;

B2 Reaction: site of light dependent reactions;

Respiration

B3 Structure: inner membrane of mitochondria;

B4 Reaction: site of oxidative phosphorylation;

B5 to increase surface area;

B6 Resp: inner mitochondria membrane highly folded/ form cristae;

B7 PS: thylakoid membrane organised into grana;

B8 PS: photosynthetic pigments - for maximum absorption of light;

B9 excite/boast electrons to a higher energy level;

B10 impermeable to ions;

B11 to allow accumulation of protons/ generation of proton gradient;

B12 hold electron carriers of ETC;

B13 to allow energy to be released during electron transfer down ETC;

B14 to harness energy released to pump H^+;

B15 hold stalked particles / ATP synthase;

B16 ATP synthase couples the movement of H^+ down its gradient;

B17 to the phosphorylation of ADP to ATP;

(c) Compare haemoglobin with insulin in term of structure and function. [6]

Similarities

C1 Globular;;

C2 hydrophobic amino acids are buried in the interior of the globular structure whilst the hydrophilic amino acid residues are on the outside;;

C3 structure maintained by disulfide bonds, ionic bonds, hydrogen bonds and hydrophobic interactions;;

Differences

		Haemoglobin	Insulin
Structure	C4	Made up of 4 polypeptide chains: 2 α subunits, 2 β subunits;	Made up of 1 (or 2: A chain and B chain) polypeptide chain;
	C5	Quaternary structure;	Tertiary structure;
	C6	Consists of non-protein component: haem group;	Consists of protein component only;
Function	C7	Transport oxygen;	Regulate glucose metabolism;

[Total: 20]

6. (a) Describe the unique features of stem cells (zygote, embryonic and adult stem cells) and explain their normal functions in a living organism. [7]

Unique features of stem cells

A1 **unspecialised cells;**

A2 **can undergo differentiation into different cell types;**

A3 **capable of dividing indefinitely;**

A4 **undergo asymmetric division, in which one cell retains its stem cell properties and the other goes on to develop into a specialized cell;**

Zygote

A5 **totipotent;**

A6 **Function - to become any cell type in the adult body and cell of extraembryonic membranes;**

Embryonic stem cells

A7 **pluripotent;**

A8 **obtained from inner cell mass;**

A9 **Function - inner cell mass can give rise to the derivates of the three germ layers: endoderm, mesoderm, and ectoderm;**

A10 **ability to become almost any kind of cell in the body to produce an adult organism except the extraembryonic membrane;**

A11 **e.g. umbilical cord stem cells can develop into white blood cells, red blood cells and platelet in the umbilical cord;**

Adult stem cells

A12 **multipotent;**

A13 **undifferentiated cell found among differentiated cells in a tissue or organ;**

A14 **Function - to maintain and repair the tissue in which they are found;**

A15 **e.g.: Hematopoietic stem cells; give rise to all the types of blood cells: red blood cells, B lymphocytes, T lymphocytes, natural killer cells, neutrophils, basophils, eosinophils, monocytes, macrophages, and platelets;**

A16 **can also be unipotent;**

A17 **e.g. epidermal stem cells give rise to skin cells;**

(b) Describe the behaviour of DNA during the nuclear division process which takes place in the stem cells. [8]

Nuclear Division: Mitosis

B1 **a process by which a cell nucleus divides to produce two daughter nuclei; containing the same chromosome number; and the same genetic makeup as the parent cell;**

Interphase

B2 **DNA molecules undergoes semi-conservative; replication;**

B3 **synthesis of 2 genetically identical daughter DNA molecules;**

Prophase

B4 **Chromatin coil around histone proteins;**

B5 **Condensing; into discrete chromosomes;**

B6 **Each chromosome now appears as 2 identical; sister chromatids;**

B7 **Joined together at the centromere;**

Metaphase

B8 **Chromosomes align at the metaphase plate;**

B9 **Kinetochore microtubules attach sister chromatids to opposite poles of cell;**

Anaphase

B10 **Causing sister chromatids of each chromosome to separate;**

Reject: split

B11 **At centromeric region;**

B12 **And move towards opposite poles of the cell;**

Reject: opposite ends/ sides

B13 **Centromeres leading;**

B14 **Each sister chromatid now known as a daughter chromosome;**

Telophase

B15 **Two sets of daughter chromosomes arrive at the poles of the cell;**

B16 **Chromosomes decondense into chromatin;**

6. **(c)** Explain the effects of pH and temperature on the rate of enzyme-catalysed reactions. [5]

(I) pH

At optimum pH, the rate of an enzyme catalysed reaction is at its maximum:

C1 **the intramolecular bonds that maintain the secondary and tertiary structures of the enzyme molecule are intact;**

C2 **the conformation of the active site is most ideal for binding of the substrate;**

At pH above or below the optimum:

C3 **changes in pH alter the charges in the acidic and basic R groups of the enzyme molecule;**

C4 **disrupts ionic bonding that helps to maintain the specific shape of the enzyme;**

C5 **the active site no longer fits the substrate;**

(II) Temperature

As temperature increases, the rate of reaction increases:

C6 **there is an increase in the kinetic energy of the enzyme and substrate molecules;**

C7 **results in an increase in the frequency of effective collisions between enzyme and substrate molecules;**

C8 **more enzyme-substrate complexes are formed per unit time;**

If the temperature is increased beyond the optimum temperature, the rate of reaction decreases rapidly:

C9 **the atoms which make up the enzyme molecule vibrate vigorously;**

C10 **the bonds (esp. hydrogen bonds and hydrophobic interactions) holding the enzyme molecule in its precise shape begin to break;**

C11 **the three-dimensional shape of the enzyme is altered / denatured;**

C12 **its active site no longer fits the substrate;**

[Total: 20]